ABSOLUTE VALUE AND EXPONENTIAL FUNCTIONS

Absolute Value and Exponential Functions Performance Task

TEACHER EDITION

Form A

Answer key + scoring guidance

DIRECTIONS 5 items • 8 points • 45 minutes • Form A

Read the scenario, define quantities, show mathematical work, and explain conclusions. The task is scored with an eight-point rubric.

Equipment Inspection Model Comparison

Form A. The equipment inspection model comparison uses $D(t) = |t - 6| + 2$ for a temperature-deviation score and $V(t) = 1200(0.8)^t$ for a dollar-value model, where t is time in years.

1. Identify the vertex, minimum, symmetry, domain, and range of D.

Scoring guidance: Vertex (6,2); minimum 2; axis $t = 6$; domain all modeled t values; range $D \geq 2$.

2. Classify V as growth or decay and state its initial value and percent rate.

Scoring guidance: V shows 20% decay; initial value $V(0) = \$1200$.

3. Evaluate both functions at $t = 3$, with correct units.

Scoring guidance: $D(3) = 5$ score units; $V(3) = \$614.40$.

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4. Compare the graphs and rates of change over the modeled domain. Explain why one has a constant multiplicative rate and the other does not.

> **Scoring guidance:** V changes by a constant factor each year. D is piecewise linear with slopes -1 and +1 around its vertex.

5. Defend which model is appropriate for deviation and which is appropriate for value. Address why their output units should not be directly compared.

> **Scoring guidance:** Absolute value models distance from a target; exponential models repeated percent change. The numerical outputs cannot be compared as the same quantity because their units differ.

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Eight-Point Rubric

Category	Points	Full-credit evidence
Modeling and setup	**2**	Defines quantities and creates an appropriate mathematical model.
Accuracy	**2**	Uses correct procedures and obtains accurate results.
Representation	**2**	Uses and labels an appropriate graph, table, equation, or structural representation.
Interpretation and justification	**2**	Explains the conclusion, units, constraints, and reasonableness.

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